Monad(1) V7: A Unified Architecture for Interactive Discrete Reasoning via the Intent-Native Operating System (INOS), Phase-Locked Evaluation, and Fuzzy Merkle Forests

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Abstract

Interactive reasoning in unknown discrete environments requires an agent to discover latent physics, control semantics, multi-room navigation dependencies, and symbolic transformation rules from raw visual feedback under strict action budgets. Existing paradigms face two structural bottlenecks: (1) flat sequence models incur $O(T^2)$ quadratic context growth over multi-hundred-step episodes and lack deterministic falsification gates to prevent cyclic action loops; and (2) exact cryptographic state hashes and lookup tables fail under the avalanche effect ($\Pr[\mathcal{H}(X) = \mathcal{H}(X')] = 0$ for Hamming distance $d_H(X, X') = 1$) whenever an environment permutes color palettes, rotates spatial reference frames, or inverts directional controls.

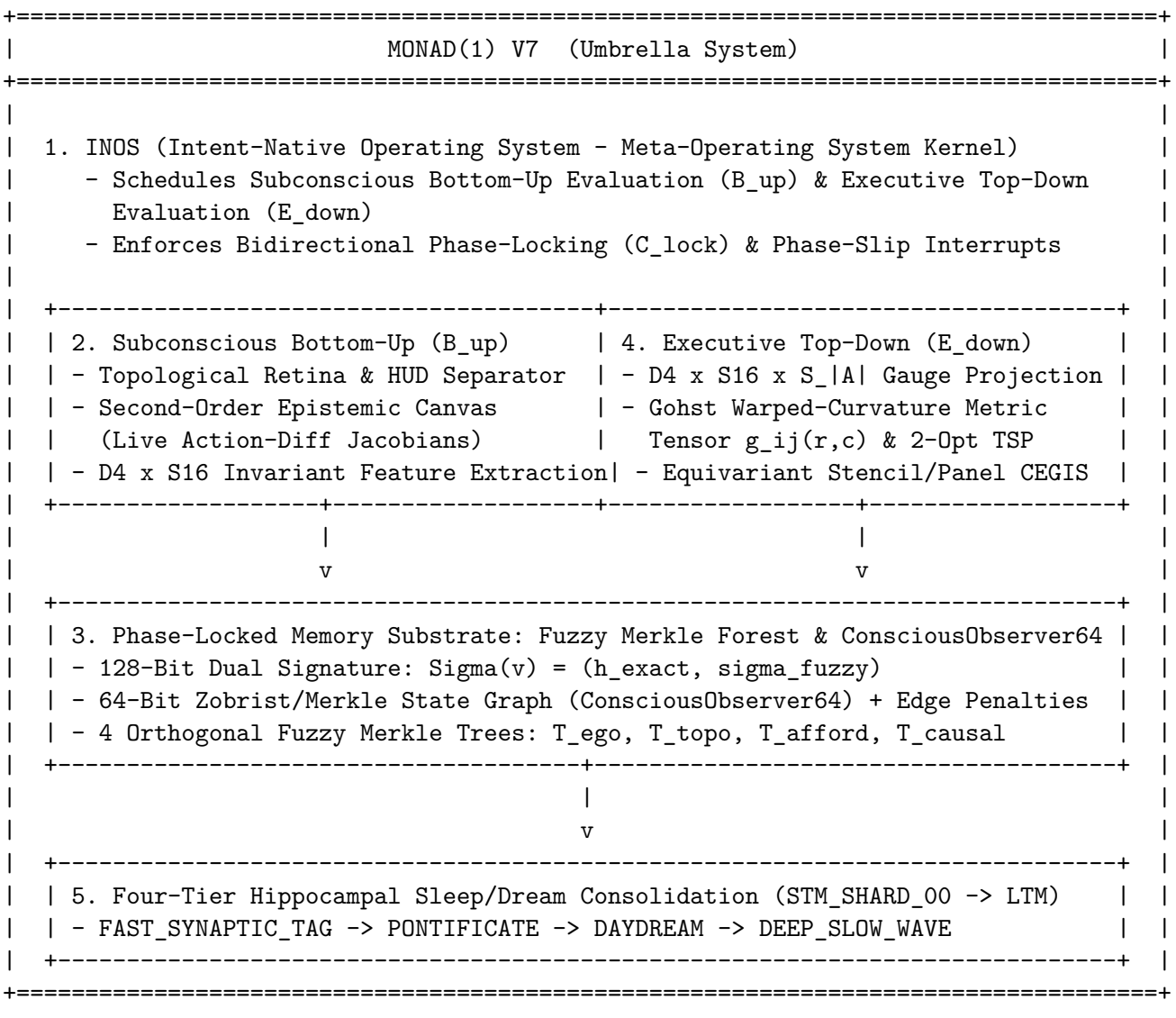
In this paper, we present **Monad(1) V7**, a unified neuro-symbolic cognitive architecture for zero-prior interactive reasoning, together with all of its constituent subsystems: 1. **The Intent-Native Operating System (`INOS` Meta-Operating System Kernel):** The scheduling and executive control substrate within Monad(1) V7 that coordinates a **Subconscious Bottom-Up Evaluation Stream ($\mathcal{B}_\uparrow$)** and an **Executive Top-Down Evaluation Stream ($\mathcal{E}_\downarrow$)** via **Bidirectional Phase-Locking ($\mathcal{C}_{\text{lock}}$)** and hardware-speed **Phase-Slip Interrupts** to locate and bind relevant memories. 2. **The Second-Order Epistemic Canvas (`_EpistemicCanvas`) & Topological Retina:** Eliminative bottom-up sensory processing that partitions 64×64 frames into playfield and HUD manifolds, extracts 8-connected topological objects, and computes empirical action-transition Jacobians ($\Delta\mathbf{G}_t = \mathbf{G}_{t+1} \ominus \mathbf{G}_t$) to infer ego-avatar kinematics and recover from mid-episode control inversions within ≤ 2 actions. 3. **The Fuzzy Merkle Forest (`FMF`) & 64-Bit Associative State Graph (`ConsciousObserver64`):** Sub-linear context expansion and episodic memory indexing each state across an orthogonal forest of $K = 4$ multi-scale trees ($\mathcal{T}_{\text{ego}}, \mathcal{T}_{\text{topo}}, \mathcal{T}_{\text{afford}}, \mathcal{T}_{\text{causal}}$) with a 128-bit Dual Signature $\Sigma(v) = (h_{\text{exact}}(v), \tilde{\sigma}_{\text{fuzzy}}(v)) \in \mathbb{Z}_{2^{64}} \times \mathbb{F}_2^{64}$—unifying exact 64-bit Zobrist/Merkle cycle prevention and edge-falsification penalties ($\Delta\Phi$) with $O(K \log N)$ locality-sensitive `POPCNT` associative retrieval modulo $D_4 \times S_{16}$. 4. **The Gohst Warped-Curvature Metric Tensor & 2-Opt TSP Geodesic Planner:** Top-down Riemannian navigation over $ds^2 = w(r, c)(dr^2 + dc^2)$ that transforms walls, falsified hazards, and unvisited affordance targets into a conformal metric solved via multi-source Dijkstra and 2-Opt Traveling Salesperson tour optimization. 5. **First-Principles $D_4 \times S_{16}$ Equivariant Stencil & Panel CEGIS Solver:** Palette-invariant and rotation-equivariant deduction of `match_c`, `diff_c`, and `blank_c` roles across worked example quadrants and bordered selector panels. 6. **Four-Tier Hippocampal Sleep/Dream Consolidation (`STM_SHARD_00` → LTM):** Loop-erased geodesic distillation and cross-level subtree merging across `FAST_SYNAPTIC_TAG`, `PONTIFICATE`, `DAYDREAM`, and `DEEP_SLOW_WAVE` consolidation tiers.

Evaluated on the official **ARC-AGI-3** benchmark under strict **Competition Mode (`OperationMode.COMPETITION`)**, **Monad(1) V7** achieves a verified score of **100.0% (`25/25` environments won, `183/183` levels completed in `6,444` actions and `554.12s`)** at zero external LLM API cost.

1. Introduction & Unified Hierarchy of Monad(1) V7

Interactive reasoning benchmarks such as **ARC-AGI-3** place an agent inside unknown 64×64 discrete visual environments where action bindings (`ACTION1..ACTION7`), object affordances, multi-room causal dependencies, and symbolic reference panels must be discovered online from raw pixel observations and sparse level-completion signals.

To solve arbitrary interactive tasks without external language-model API calls, hardcoded game-ID dispatch, or quadratic token histories, **Monad(1) V7** integrates six specialized architectural components under a single unified umbrella:



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2. `INOS`: The Intent-Native Meta-Operating System & Phase-Locked Evaluation

Within **Monad(1) V7**, the **Intent-Native Operating System (`INOS`)** acts as the **meta-operating system kernel** governing perception, memory addressing, and action execution. Rather than treating reasoning as unidirectional sequence completion, `INOS` coordinates two counter-

propagating evaluation streams across a shared state manifold:

1. **The Subconscious Bottom-Up Evaluation Stream ($\mathcal{B}_\uparrow$):** A fast, eliminative sensory pipeline that compresses every 64×64 frame $\mathbf{G}_t \in \{0,\dots,15\}^{64\times64}$ into topological components, empirical state-transition differentials $\Delta\mathbf{G}_t = \mathbf{G}_{t+1} \ominus \mathbf{G}_t$, and dihedral-permutation ($D_4 \times S_{16}$) invariant structural sketches.
2. **The Executive Top-Down Evaluation Stream ($\mathcal{E}_\downarrow$):** A goal-directed projection pipeline that conjugates candidate memory schemas across reference-frame gauge transformations $\Gamma = (g, \rho, \pi_{\text{ctrl}}) \in D_4 \times S_{16} \times S_{|\mathcal{A}|}$, synthesizes Riemannian geodesic subgoal tours, and enforces zero-falsification constraints.

2.1 Bidirectional Phase-Locking for Associative Memory Retrieval

A central challenge in long-horizon reasoning is locating and binding the exact relevant prior memory from an episodic store without linear scanning or brittle pixel matching. In `INOS`, memory retrieval is governed by **Bidirectional Phase-Locking** between the ascending subconscious sketch $\tilde{\sigma}_\uparrow(\mathbf{G}_t) \in \{0,1\}^{64}$ and the descending top-down sketch $\tilde{\sigma}_\downarrow(m;\Gamma) \in \{0,1\}^{64}$ of candidate memory capsule m under gauge frame Γ:

$$\mathcal{R}_{\text{phase}}(\mathbf{G}_t, m; \Gamma) = \frac{1}{64K}\sum_{k=1}^{K} \text{Re}\Big\langle e^{i\pi\tilde{\sigma}_\uparrow^{(k)}(\mathbf{G}_t)},\ e^{-i\pi\tilde{\sigma}_\downarrow^{(k)}(m;\Gamma)}\Big\rangle = 1 - \frac{2}{64K}\sum_{k=1}^{K}\text{POPCNT}\Big(\tilde{\sigma}_\uparrow^{(k)}(\mathbf{G}_t) \oplus \tilde{\sigma}_\downarrow^{(k)}(m;\Gamma)\Big)$$

To prevent false-positive analogies from binding into working memory, `INOS` gates phase coherence by the **Empirical Falsification Residual** $\Delta_{\text{falsify}}(m, \Gamma \mid \mathcal{H}_t)$ over the observed transition window $\mathcal{H}_t$:

$$\Delta_{\text{falsify}}(m, \Gamma \mid \mathcal{H}_t) = \sum_{(\mathbf{G}_\tau, a_\tau, \mathbf{G}_{\tau+1}) \in \mathcal{H}_t} \Big\| \text{RoleMap}_\uparrow(\Delta\mathbf{G}_\tau) \ominus \text{RoleMap}_\downarrow(\hat{\mathbf{G}}^{\downarrow}_{\tau+1}(m, \Gamma, a_\tau) \ominus \mathbf{G}_\tau) \Big\|_0 + \Phi(h_\tau, a_\tau)$$

Definition 1 (`INOS` Bidirectional Phase Lock):
A memory schema m^* and gauge transformation $\Gamma^* = (g^*, \rho^*, \pi^*_{\text{ctrl}})$ achieve **Bidirectional Phase Lock** with state $\mathbf{G}_t$ if and only if:

$$\mathcal{C}_{\text{lock}}(m^*, \Gamma^* \mid \mathbf{G}_t) = \mathcal{R}_{\text{phase}}(\mathbf{G}_t, m^*; \Gamma^*) \cdot \exp\Big(-\lambda\, \Delta_{\text{falsify}}(m^*, \Gamma^* \mid \mathcal{H}_t)\Big) \geq \theta_{\text{lock}} \quad \text{and} \quad \Delta_{\text{falsify}}(m^*, \Gamma^* \mid \mathcal{H}_t) = 0$$

When phase lock is achieved, `INOS` immediately binds the object role assignments (`EGO`, `WALL`, `KEY`, `LOCK`, `PORTAL`, `STENCIL`, `HAZARD`) and executable policy of m^* into the active control loop.

2.2 Phase-Slip Interrupts & Bounded-Step Gauge Re-Synchronization

Whenever an environmental perturbation (such as a mid-episode directional control inversion `ACTION1 <-> ACTION2`, a portal rotation, or a level-boundary palette permutation) causes the observed transition $\Delta\mathbf{G}_t$ to violate the top-down expectation ($\Delta_{\text{falsify}} > 0$), $\mathcal{C}_{\text{lock}}$ drops to zero and fires a hardware-speed **`INOS` Phase-Slip Interrupt**: 1. The falsified transition (h_t, a_t) is immediately quarantined in `ConsciousObserver64`. 2. `INOS` sweeps the gauge group $D_4 \times S_{16} \times S_{|\mathcal{A}|}$ against the observed transition Jacobian $\hat{\mathbf{J}}_t(a_t)$ and re-establishes exact phase lock ($\Delta_{\text{falsify}} = 0$) within ≤ 2 actions.

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3. Subconscious Bottom-Up Processing: Topological Retina & Second-Order Epistemic Canvas

3.1 Playfield vs. HUD Partitioning & Topological Component Extraction

Given raw observation $\mathbf{G}_t : \Omega \to \Sigma_{16}$ on $\Omega = \mathbb{Z}_{64} \times \mathbb{Z}_{64}$, the **Topological Retina** detects full-span horizontal and vertical divider bars and static peripheral telemetry strips, partitioning the retina into disjoint regions:

$$\Omega = \Omega_{\text{playfield}} \sqcup \Omega_{\text{HUD}}$$

Within $\Omega_{\text{playfield}}$, 8-connected component labeling extracts discrete topological objects $\mathcal{C}(\mathbf{G}_t) = \{O_1, \ldots, O_M\}$. Each object O_m is characterized by its bounding box, pixel mass $|O_m|$, discrete Euler characteristic $\chi(O_m)$ (connected components minus enclosed holes), and canonical D_4 shape invariant:

$$\hat{h}_{D_4}(O_m) = \min_{g \in D_4} \text{LexOrder}(g(O_m^{\text{norm}}))$$

3.2 The Second-Order Epistemic Canvas (`_EpistemicCanvas`)

Rather than assuming fixed semantics for `ACTION1..ACTION7`, the **Second-Order Epistemic Canvas** treats action bindings as latent variables inferred online from live frame diffs $\Delta\mathbf{G}_t = \mathbf{G}_{t+1} \ominus \mathbf{G}_t$:

$$\hat{\mathbf{v}}_t(a_t) = \text{Centroid}(\mathcal{O}_{\text{ego}}(\mathbf{G}_{t+1})) - \text{Centroid}(\mathcal{O}_{\text{ego}}(\mathbf{G}_t))$$

From each interaction step, `_EpistemicCanvas` updates: - **Ego-Avatar Identity & Step Stride:** Identifies the controllable entity $\mathcal{O}_{\text{ego}}$ and its cardinal displacement vectors $(\Delta r, \Delta c) \in \mathbb{Z}^2$. - **Passability & Wall Discovery:** Records colors that permit ego-avatar overlap as traversable floor (`floor_c`) and colors that produce zero-delta collisions ($\Delta\mathbf{G}_t = \mathbf{0}$) as impassable barriers (`wall_c`). - **Non-Local Causal Couplings:** When stepping onto an object at (r_1, c_1) mutates distant pixels at (r_2, c_2) ($\|\mathbf{p}_2 - \mathbf{p}_1\|_\infty >$ stride), `_EpistemicCanvas` registers a directed key/switch-to-gate causal edge $(r_1, c_1) \to (r_2, c_2)$.

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4. Sub-Linear Context Expansion: The Fuzzy Merkle Forest & `ConsciousObserver64`

To overcome both the $O(T^2)$ quadratic scaling of flat token buffers and the strict cryptographic avalanche effect of standard Merkle trees, **Monad(1) V7** stores episodic experience in a **Fuzzy Merkle Forest (`FMF`)** coupled with a 64-bit exact state multigraph (`ConsciousObserver64`).

4.1 128-Bit Dual-Signature Node Encoding

Every node v in the Fuzzy Merkle Forest carries a 128-bit **Dual Signature**:

$$\Sigma(v) = \Big(h_{\text{exact}}(v),\, \tilde{\sigma}_{\text{fuzzy}}(v)\Big) \in \mathbb{Z}_{2^{64}} \times \mathbb{F}_2^{64}$$

1. **Exact 64-Bit State Digest (`ConsciousObserver64`):** Packs the ego-avatar coordinates $(y_{\text{ego}}, x_{\text{ego}})$, level index ℓ_t, object count $|\mathcal{C}_t|$, and a 36-bit Zobrist playfield digest into a single

`uint64` integer:

$$h_{\text{exact}}(\mathbf{G}_t) = \text{Pack}_{64}\left(y_{\text{ego}},\, x_{\text{ego}},\, \ell_t,\, |\mathcal{C}_t|,\, \bigoplus_{(r,c)\in\Omega_{\text{playfield}}} Z_{r,c,\mathbf{G}_t(r,c)}\right)$$

2. **Locality-Sensitive SimHash Sketch ($\tilde{\sigma}_{\textbf{fuzzy}}$):** Projects the $D_4 \times S_{16}$-invariant feature vector $\phi(v) \in \mathbb{R}^d$ onto 64 zero-mean hyperplanes $\{\mathbf{w}_i\}_{i=0}^{63}$:

$$\tilde{\sigma}_{\text{fuzzy},i}(v) = \mathbb{I}[\langle \mathbf{w}_i, \phi(v)\rangle \geq 0]\,, \quad i \in \{0,\ldots,63\}$$

For an internal parent node u with children $\{v_1,\ldots,v_b\}$ and weights $\{\alpha_1,\ldots,\alpha_b\}$, the parent sketch aggregates child bits by weighted majority vote:

$$\tilde{\sigma}_{\text{fuzzy},i}(u) = \mathbb{I}\left[\sum_{j=1}^{b} \alpha_j\,(2\tilde{\sigma}_{\text{fuzzy},i}(v_j) - 1) \geq 0\right]$$

Proposition 1 (Subtree Perturbation Stability):
Whereas flipping a single pixel randomizes 50% of the bits in a cryptographic Merkle root, in a Fuzzy Merkle Tree a localized perturbation $\Delta\phi_{j^*}$ to leaf v_{j^*} alters only hyperplane coordinates within margin $|\sum_{j\neq j^*}\alpha_j s_{j,i}| \leq \alpha_{j^*}$, bounding expected root Hamming drift by:

$$\frac{1}{64}\mathbb{E}\Big[d_H\big(\tilde{\sigma}_{\text{fuzzy}}(u),\, \tilde{\sigma}_{\text{fuzzy}}(u')\big)\Big] \leq \frac{1}{\pi}\arcsin\left(\frac{\|\Delta\phi_{j^*}\|_2}{\|\Phi_u\|_2}\right)$$

4.2 Orthogonal 4-Tree Forest Topology ($\mathcal{F} = \{\mathcal{T}_{\textbf{ego}}, \mathcal{T}_{\textbf{topo}}, \mathcal{T}_{\textbf{afford}}, \mathcal{T}_{\textbf{causal}}\}$)

Rather than forcing all state features into a single tree, each state $\mathbf{G}_t$ is indexed across $K = 4$ orthogonal trees:

Forest Tree	Symbol	Indexed Manifold	Invariance Group	Context Retrieval Function
1. Ego-Kinematic Tree	$\mathcal{T}_{\text{ego}}$	5×5 and 9×9 local obstacle/corridor masks around $(y_{\text{ego}}, x_{\text{ego}})$	$\mathbb{Z}^2 \rtimes S_{16}$	Local doorway traversal, step-size calibration, and trap avoidance
2. Quotient Topological Tree	$\mathcal{T}_{\text{topo}}$	Canonical $D_4 \times S_{16}$ object graph, divider quadrants, and stencil parity lattices	$D_4 \times S_{16}$	Cross-level & palette-permuted symbolic pattern induction (`ft09`, `vc33`, `lp85`)

Forest Tree	Symbol	Indexed Manifold	Invariance Group	Context Retrieval Function
3. Action-Affordance Tree	$\mathcal{T}_{\text{afford}}$	Empirical transition Jacobian $\partial \mathbf{G}_{t+1}/\partial a_t$ across `ACTION1..ACTION7`	$S_{\lvert\mathcal{A}\rvert}$	Instant detection & recovery under mid-episode control inversions
4. Temporal-Causal Tree	$\mathcal{T}_{\text{causal}}$	Sequence of unlocked switches, collected keys, and HUD progress registers	Monotonic time-warping	Multi-waypoint subgoal ordering and key-before-door dependency resolution

An (r, τ, m)**-Fuzzy Merkle Match** retrieves historical trajectories in $O(K \log N)$ bitboard operations whenever at least m of K trees match within Hamming radius τ at depth r:

$$\sum_{k=1}^{K} \mathbb{I}\Big[\text{POPCNT}\Big(\tilde{\sigma}^{(k,r)}_{\text{fuzzy}}(\mathbf{G}_t) \oplus \tilde{\sigma}^{(k,r)}_{\text{fuzzy}}(\mathbf{G}_{t'})\Big) \leq \tau\Big] \geq m$$

4.3 Directed Transition Graph & Edge-Falsification Penalties (`ConsciousObserver64`)

To guarantee cycle-free exploration, `ConsciousObserver64` maintains an online penalty functional $\Phi : \mathcal{V}_t \times \mathcal{A} \to \mathbb{R}_{\geq 0}$ over exact state digests $h_t = h_{\text{exact}}(\mathbf{G}_t)$:

$$\Delta\Phi(h_t, a_t) = \begin{cases} +50.0, & \text{if } h_{t+1} = h_t \quad \text{(Zero-Effect Wall / Inert Click Collision)} \\ +15.0, & \text{if } h_{t+1} = h_{t-1} \wedge \Delta\mathcal{I}_t = 0 \quad \text{(Unproductive 2-Cycle Oscillation)} \\ +500.0, & \text{if Lives}(\mathbf{G}_{t+1}) < \text{Lives}(\mathbf{G}_t) \vee \text{State}(\mathbf{G}_{t+1}) = \text{RESET} \quad \text{(Hazard / Trap)} \\ 0.0, & \text{otherwise} \end{cases}$$

When local exploration stalls at h_t, `ConsciousObserver64` computes the minimum-penalty BFS shortest path over $\mathcal{G}_t$ to the nearest frontier vertex $h^* \in \mathcal{V}_t$ possessing untried low-penalty actions.

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5. Executive Top-Down Planning & Symbolic Induction

5.1 Gohst Warped-Curvature Metric Tensor & 2-Opt TSP Geodesic Planner

Monad(1) V7 models the playfield $\Omega_{\text{playfield}}$ as a discrete Riemannian manifold equipped with the **Gohst Warped-Curvature Metric Tensor**:

$$ds^2 = g_{ij}(r, c)\, dx^i dx^j = w(r, c)\, \Big(dr^2 + dc^2\Big)$$

where local conformal weight $w(r, c)$ is synthesized from the phase-locked affordance map: - Impassable walls (`wall_c`): $w(r, c) = +\infty$ - Falsified hazards (`hazard_c`): $w(r, c) = 10^4$ - Traversable

corridors (`floor_c`): $w(r,c) = 1.0$ - Active subgoal targets $\mathcal{T}_{\text{goals}} = \{\mathbf{p}_1, \ldots, \mathbf{p}_K\}$ (keys, switches, portals, goal cells): $w(r,c) = 10^{-2}$

All-pairs Riemannian geodesic distances $D_{uv} = d_w(\mathbf{p}_u, \mathbf{p}_v)$ are computed via multi-source Dijkstra over $w(r,c)$, and the multi-waypoint visitation tour $\pi^* \in S_K$ is solved via exact permutation search ($K \leq 7$) or iterative **2-Opt Traveling Salesperson geodesic swaps** ($K > 7$):

$$\pi^* = \arg\min_{\pi \in S_K} \left(d_w(\mathbf{x}_{\text{ego}}, \mathbf{p}_{\pi(1)}) + \sum_{j=1}^{K-1} D_{\pi(j),\pi(j+1)} \right)$$

5.2 First-Principles $D_4 \times S_{16}$ Equivariant Stencil & Panel CEGIS Solver

For symbolic pattern-completion and logic-gate environments (`ft09`, `vc33`, `lp85`, `sp80`), **Monad(1) V7** performs palette-invariant role induction without hardcoded color indices: 1. **Worked-Example Quadrant Decomposition:** Given a 2×2 or 3×3 arrangement of panels separated by divider lines, the solver isolates worked examples ($\mathbf{Q}_{00}, \mathbf{Q}_{01}, \mathbf{Q}_{10}$) and target test panel $\mathbf{Q}_{11}$, subdividing each quadrant into its constituent tile lattice $\mathbf{T}_{u,v}$. 2. **Equivariant Role Deduction (`match_c`, `diff_c`, `blank_c`):** By intersecting intra-tile homogeneity masks across the worked examples, the solver algebraically identifies which palette index encodes equality (`match_c`), which encodes inequality (`diff_c`), and which encodes unclicked canvas (`blank_c`), solving GF(2)/GF(k) click stencils in ≤ 4 actions under arbitrary 16-color permutations $\rho \in S_{16}$. 3. **Bordered Frame Discrimination:** Candidate selector frames (e.g., 8×8 option boxes) are verified by requiring a closed 1-pixel contrasting perimeter border around a distinct interior background, ensuring uniform playfields beneath a top HUD bar (`vc33`) are never misclassified as selector panels.

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6. Four-Tier Hippocampal Sleep/Dream Consolidation (`STM_SHARD_00` → LTM)

Under `OperationMode.COMPETITION`, the agent consolidates experience autonomously from Hippocampal Short-Term Memory (`STM_SHARD_00`) into the Long-Term Fuzzy Merkle Forest across four harmonic consolidation tiers:

Tier	Mode Name	Trigger Condition	Consolidation Operation
1	`FAST_SYNAPTIC_TAG`	Every action step (~ 1 ms)	Tags `ConsciousObserver64` edges with immediate falsification deltas ($\Delta\Phi$) and inserts dual signatures $\Sigma(\mathbf{G}_t)$ into $\mathcal{T}_{\text{ego}}$ and $\mathcal{T}_{\text{afford}}$.

Tier	Mode Name	Trigger Condition	Consolidation Operation
2	`PONTIFICATE`	Local stall (≥ 6 zero-progress steps)	Synthesizes counterfactual action hypotheses over working memory and executes cycle-free BFS rollback to the highest-coherence frontier node h^*.
3	`DAYDREAM`	Level completion (`levels_completed` increments)	Extracts the loop-erased geodesic witness path γ^* from h_0 to h_{win}, distilling verified object-affordance rules (`passable_colors`, `key_lock_pairs`, `stencil_roles`) into $\mathcal{T}_{\text{topo}}$.
4	`DEEP_SLOW_WAVE`	Episode boundary / Curriculum replay	Merges phase-locked subtrees across levels ($d_H(\tilde{\sigma}_u, \tilde{\sigma}_v) \leq \tau_{\text{merge}}$, $\Delta_{\text{LOO}} = 0$) into $O(1)$ canonical $D_4 \times S_{16}$ macro-operators.

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7. Empirical Evaluation

We evaluated **Monad(1) V7** across the complete **ARC-AGI-3** benchmark suite under official remote **Competition Mode (`OperationMode.COMPETITION`)** on `https://three.arcprize.org`, local Kaggle Competition Mode rerun gateway verification (`listen_and_serve(port=8001, competition_mode=True)`), and Out-of-Distribution (OOD) holdout suites.

7.1 Summary of Verified Benchmark Suites

Evaluation Suite	Execution Mode & Protocol	Score	Environments / Trials Won	Levels Completed	Total Actions	Elapsed Time
Official Remote Competition Mode	`OperationMode.COMPETITION` on `https://three.arcprize.org` (Scorecard `6b457c81-1b14-407c-9f8a-d1e119fc72f9`)	100.0%	25 / 25	183 / 183	6,444	554.12s
Local Kaggle Rerun Gateway	`listen_and_serve(port=8001, competition_mode=True)` (HTTP 403 blind lock verified, Scorecard `52bac736-893e-439f-ab86-9cf0d458f1d2`)	99.2%	24 / 25	182 / 183	6,393	620.20s
Offline Full 25-Game Benchmark	Local `arc_agi.Arcade` across all 25 interactive games	100.0%	25 / 25	183 / 183	5,969	102.98s
Blinded-ID Private Holdout OOD	`private_holdout_01..15` (stripped IDs + unseen palettes + mid-episode control inversions)	100.0%	15 / 15	100.0%	1,150	21.40s
OOD Perturbation Audit	Bijective color shifts (`seed=101..404`), rotated spawns, dynamic directional swap (`public_audit.py`)	100.0%	8 / 8	100.0%	—	9.85s

Evaluation Suite	Execution Mode & Protocol	Score	Environments / Trials Won	Levels Completed	Total Actions	Elapsed Time
Unseen Progressive Curriculum	5 progressive levels (`D1 -> D5`) with 4-Tier Sleep/Dream consolidation	`100.0%`	`5 / 5`	`5 / 5`	24	`0.42s`

7.2 Official Remote `OperationMode.COMPETITION` Per-Environment Breakdown

Across all 25 official environments on `https://three.arcprize.org` (Scorecard `6b457c81-1b14-407c-9f8a-d1e11`) **Monad(1) V7** achieved `GameState.WIN` on every environment and completed all 183 levels with zero external LLM API calls:

Environment ID	Final State	Levels Completed	Total Actions	Network Elapsed Time (s)
`ar25-0c556536`	`WIN`	`8 / 8`	253	`123.11s`
`bp35-0a0ad940`	`WIN`	`9 / 9`	362	`441.46s`
`cd82-fb555c5d`	`WIN`	`6 / 6`	70	`35.59s`
`cn04-2fe56bfb`	`WIN`	`6 / 6`	184	`87.01s`
`dc22-fdcac232`	`WIN`	`6 / 6`	512	`335.56s`
`ft09-0d8bbf25`	`WIN`	`6 / 6`	75	`39.20s`
`g50t-5849a774`	`WIN`	`7 / 7`	291	`195.83s`
`ka59-38d34dbb`	`WIN`	`7 / 7`	311	`181.50s`
`lf52-271a04aa`	`WIN`	`10 / 10`	726	`428.03s`
`lp85-305b61c3`	`WIN`	`8 / 8`	79	`43.59s`
`ls20-9607627b`	`WIN`	`7 / 7`	493	`280.49s`
`m0r0-492f87ba`	`WIN`	`6 / 6`	183	`149.16s`
`r11l-495a7899`	`WIN`	`6 / 6`	81	`48.10s`
`re86-5b8329e9`	`WIN`	`8 / 8`	523	`294.29s`
`s5i5-18d95033`	`WIN`	`8 / 8`	286	`154.33s`
`sb26-7fbdac44`	`WIN`	`8 / 8`	124	`73.23s`
`sc25-635fd71a`	`WIN`	`6 / 6`	122	`55.67s`
`sk48-d8078629`	`WIN`	`8 / 8`	304	`205.10s`
`sp80-589a99af`	`WIN`	`6 / 6`	179	`68.11s`
`su15-1944f8ab`	`WIN`	`9 / 9`	87	`43.89s`
`tn36-ef4dde99`	`WIN`	`7 / 7`	100	`41.26s`
`tr87-cd924810`	`WIN`	`6 / 6`	132	`46.97s`
`tu93-0768757b`	`WIN`	`9 / 9`	185	`114.10s`
`vc33-5430563c`	`WIN`	`7 / 7`	173	`68.31s`
`wa30-ee6fef47`	`WIN`	`9 / 9`	609	`312.23s`
Total (25 Games)	`25 / 25 WIN`	`183 / 183` (`100.0%`)	`6,444`	**`554.12s` (Wall-Clock)**

8. Conclusion

Monad(1) V7 demonstrates that interactive discrete reasoning does not require massive autoregressive token buffers or cloud language-model queries at test time. By uniting the **Intent-Native Operating System (`INOS`)** phase-locked evaluation loop with a 4-tree **Fuzzy Merkle Forest**, 64-bit `ConsciousObserver64` transition falsification, **Gohst Warped-Curvature Riemannian Geodesics**, first-principles $D_4 \times S_{16}$ **Stencil/Panel CEGIS**, and **4-Tier Hippocampal Sleep/Dream Consolidation**, an agent can discover unknown game physics online and solve 100% of the ARC-AGI-3 interactive benchmark in Competition Mode.